



Pyrolysis polygeneration of pine nut shell: Quality of pyrolysis products and study on the preparation of activated carbon from biochar



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HIGHLIGHTS

- Temperature has remarkable effect on the quality and energy distribution of products.
- Biochar with highest energy yield is concerned to be the primary pyrolysis product.
- 600 °C was selected as the optimum temperature for biochar production.
- Activation condition has significant effect on the performance of activated carbon.
- Activated carbon with specific surface area of 1057.8 m²/g was obtained from biochar.

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ABSTRACT

A lab-scale pyrolysis reactor was utilized to investigate the effect of pyrolysis temperature (300–700 °C) on the yield, quality, and energy distribution of products issued from the pyrolysis polygeneration of pine nut shells. Afterward, activated carbon was prepared from biochar using the steam activation method. Pyrolysis temperatures ranging from 500 to 600 °C were found to be optimal in inducing products with improved properties, such as higher heating values of non-condensable gas, lower water content and elevated heating values of bio-oil, and substantial fixed carbon content and greater specific surface area of biochar. In addition, it was noticed that the activation conditions had a significant effect on the yield and adsorption performance of the activated carbon. As a result, activated carbon with elevated specific surface area reaching 1057.8 m²/g was obtained at the optimal conditions of 850 °C activation temperature, 80 min activation time, and 1.5 steam/biochar ratio.

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1. Introduction

The shell of pine nut is a primary byproduct that accounts for approximately 40% of the total mass of pine nut. In recent years, the increase in centralized processing for pine nut raised the annual production of the shells. Pine nut shells have a substantial heating value, high fixed carbon content, low ash content, and elevated mechanical strength (Kim et al., 2014; Uçar and Karagöz, 2014). Hence, it is a good source of raw material for pyrolysis polygeneration of biomass, as well as a suitable material for the preparation of high-quality activated carbon (Deng et al., 2014). If pine nut shells are effectively utilized, this can not only save resources and protect the environment but also promote the rapid, sustainable development of the pine nut industry and improve the economic benefits.

A biomass pyrolysis polygeneration process refers to the thermochemical conversion of biomass that comprehensively utilizes the products obtained from pyrolysis like bio-oil, non-condensable gas, and biochar, for maximizing the economic and environmental benefits. The biomass pyrolysis polygeneration technology has attracted increasing attention in recent years (Chen et al., 2015a, 2012, 2014d; Ren et al., 2014; Yang et al., 2016). For example, Chen et al. have pyrolyzed cotton stalk to evaluate the effects of temperature on the characteristics of the resulting three-phase products and found that temperatures ranging from 550 to 750 °C were suitable for the production of an efficient pyrolysis (Chen et al., 2012). Other materials such as bamboo have been investigated under different temperatures to evaluate the characteristics of the resulting three-phase products, carbon, and energy yields (Chen et al., 2015a). However, previous studies had mainly focused on straw and/or wood, but nut shells have rarely been reported.

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Activated carbon is primarily prepared from substances containing high carbon content, such as agricultural wastes, sawdust, and shells (Abioye and Ani, 2015; Cagnon et al., 2009; Maneerung et al., 2016; Yahya et al., 2015). Materials based on activated carbon generally exhibited multi-porous structures, excellent adsorption efficacies, and good chemical inertness, which were widely applied in various fields like food, chemicals, pharmaceuticals, and environmental protection (Abioye and Ani, 2015; Maneerung et al., 2016). The pine nut shells are a good source of raw material for the preparation of activated carbon, especially using the steam activation process due to its simplicity, easy cleaning, and cost-effectiveness (Deng et al., 2014). Consequently, steam activation is one of the most widely used methods for the industrial preparation of activated carbon (Cagnon et al., 2009; Maneerung et al., 2016; Shim et al., 2015). The pyrolysis polygeneration of pine nut shell followed by processing of the biochar into activated carbon is, hence, a novel process for pine nut shell utilization.

This study aimed to evaluate the effect of pyrolysis temperature on the reaction yields, as well as the characteristics and energy distribution of the resulting products (bio-oil, biochar, non-condensable gas) obtained from the pyrolysis polygeneration process of pine nut shells. Activated carbon was then prepared using the biochar issued from the pyrolysis process using steam as the activator. The effects of the activation temperature, activation time, and amount of steam on the performance of the resulting activated carbon were also examined.

2. Materials and methods

2.1. Materials

Pine nut shells obtained from Liaoyuan city in Jilin province of China were used as materials and placed in a furnace at 105 °C for 24 h before the experiments.

2.2. Pyrolysis experiment

The pyrolysis processes were performed at temperatures of 300, 400, 500, 600, and 700 °C using a lab-scale fixed bed pyrolysis reactor used in the previous study (Chen et al., 2015b). Samples of about 50 g in mass were used for each experiment at a nitrogen flow rate of 400 mL/min. Silicawool with stainless steel wires were employed as supports for the samples, as well as to enhance the heat transfer effect and limit heat and mass transmission gradients in the samples. A thermometer was inserted within the samples to measure the real temperature. The samples were heated rapidly from room to the target temperature, maintained for 10 min, and then the biochar was collected for further analysis. The weight of the condenser pipe was weighted before and after the experiments, and the difference indicated the amount of the produced bio-oil. The non-condensable gas was collected in a gas bag. As it was hard to measure, its yield was determined by difference according to previously published studies (Chen et al., 2015b; Uçar and Karagöz, 2014).

2.3. Preparation of activated carbon

A small activation furnace was employed to prepare the activated carbon from biochar issued from pine nut shells by placing a mass of approximately 15 g biochar in the activation furnace. After the removal of air by purged nitrogen in the furnace, the temperature was increased inside the unit to the pre-set temperature under air-free conditions. High-purity nitrogen was then passed along with the steam through a quartz tube to produce a reaction with the biochar. After a certain time, the inlet steam was cut off,

and the reaction system was cooled to room temperature to obtain the activated carbon. The following parameters were utilized in this study: activation temperatures of 750–900 °C, activation times of 40–120 min, and steam/biochar ratios of 0.5–2.5.

2.4. Products characterization

2.4.1. Analysis of liquid products

The bio-oil obtained from the pyrolysis of pine nut shells was at first homogeneous but after several days of storage, a phase separation consisted of an aqueous and oil phases occurred in the bio-oil. Although it was not clear when the exact process of separation took place, a distinct separation line between the water and oil phases appeared six days after the storage in a refrigerator at 5 °C. A funnel was used to separate between both phases, then the mass of the oil phase was determined by weighting.

The physicochemical properties of the oil phase were determined by several analytical methods. The ultimate analysis was evaluated by an elemental analyzer (Vario macro cube, Elementar, Germany) and oxygen content was determined by difference. The water content of the bio-oil and its oil phase were analyzed by Karl-Fischer titration. The organic components in the oil phase were examined by gas chromatography coupled to a mass spectrometer (GC/MS 7890A/5975C, Agilent Company, USA), and the obtained results were matched using a reference standard library GG1034C software. The higher heating value (HHV) of the bio-oil and its oil phase were estimated in an adiabatic oxygen bomb calorimeter (XRY-1A, Changji Geological Instruments, China).

2.4.2. Analysis of gas products

The non-condensable gas obtained from the pyrolysis of pine nut shells were collected in a gas bag then analyzed by a gas chromatography analyzer (GC-TCD 7890 II, Shanghai Tianmei, China). The HHV of the non-condensable gas was obtained by summing each gas concentration with its higher heating value following Eq. (1):

$$\begin{aligned} \text{HHV}_{\text{gas}}(\text{MJ}/\text{Nm}^3) = & \text{Vol}_{\text{CO}}(\%) \times \text{HHV}_{\text{CO}}(\text{MJ}/\text{Nm}^3) \\ & + \text{Vol}_{\text{H}_2}(\%) \times \text{HHV}_{\text{H}_2}(\text{MJ}/\text{Nm}^3) \\ & + \text{Vol}_{\text{CH}_4}(\%) \times \text{HHV}_{\text{CH}_4}(\text{MJ}/\text{Nm}^3) \\ & + \text{Vol}_{\text{C}_2+}(\%) \times \text{HHV}_{\text{C}_2+}(\text{MJ}/\text{Nm}^3). \end{aligned} \quad (1)$$

The overall gas density can be calculated from the density of each gas.

2.4.3. Analysis of solid products

The proximate analyses of pine nut shells, biochar, and activated carbon were performed according to the Chinese National Standards GB/T28731-2012 (Proximate analysis of solid biofuels). The contents of hemicellulose, cellulose, and lignin in the pine nut shells were resolved using the modified Van Soest method (Chen et al., 2014c). The utilized HHV analysis methods of pine nut shells, biochar, and activated carbon were similar to those of the liquid products. The adsorption capacity of the activated carbon was evaluated according to the Chinese National Standards GB/T 12496.8-2015 (Test Methods of wooden activated carbon – Determination of iodine number). The specific surface area and pore size of both the biochar and activated carbon were characterized by an ASIQM0002-2 system (US Quantachrome Instruments). The BET method was also used to estimate the specific surface area of the products. The analyses of all the products characterizations were replicated three times. The average of data and the standard deviation were listed in the Tables 1–4.

Table 1

Characteristics of the oil phase and bio-oil obtained at different pyrolysis temperatures.

Temperature (°C)	Ultimate analysis of oil phase (wt%)				Water content (wt%)		HHV (MJ/kg)		Mass fraction of oil phase (wt%)
	C	H	N	S	Bio-oil	Oil phase	Bio-oil	Oil phase	
300	57.85 ± 0.83	5.95 ± 0.21	0.32 ± 0.04	0.15 ± 0.01	75.32 ± 0.89	8.69 ± 0.31	7.11 ± 0.26	22.69 ± 0.52	17.36 ± 0.82
400	58.43 ± 0.81	7.27 ± 0.18	0.29 ± 0.02	0.16 ± 0.01	61.56 ± 0.85	7.67 ± 0.39	13.87 ± 0.43	23.67 ± 0.59	32.62 ± 1.03
500	62.39 ± 0.55	7.35 ± 0.17	0.25 ± 0.03	0.13 ± 0.01	53.37 ± 0.73	7.12 ± 0.43	15.76 ± 0.45	25.12 ± 0.45	46.71 ± 1.25
600	63.18 ± 0.62	7.13 ± 0.19	0.25 ± 0.03	0.19 ± 0.01	55.62 ± 0.46	7.63 ± 0.35	15.12 ± 0.37	25.63 ± 0.56	45.25 ± 1.09
700	63.73 ± 0.79	7.09 ± 0.15	0.21 ± 0.02	0.17 ± 0.01	56.21 ± 0.61	7.87 ± 0.38	15.85 ± 0.62	25.87 ± 0.51	45.73 ± 1.27

Table 2

The identified compounds in the oil phase separated from the bio-oil obtained at different temperatures.

No.	Compounds	Pear area %			
		400 °C	500 °C	600 °C	700 °C
Phenols					
1	Phenol	5.87 ± 0.35	7.32 ± 0.23	7.05 ± 0.39	7.96 ± 0.51
2	2-Methylphenol	1.59 ± 0.03	1.16 ± 0.08	1.23 ± 0.13	1.82 ± 0.06
3	4-Methylphenol	3.56 ± 0.32	2.13 ± 0.51	5.37 ± 0.33	3.28 ± 0.31
4	2-Methoxyphenol	6.67 ± 0.27	8.21 ± 0.83	6.65 ± 0.81	7.12 ± 0.52
5	2,4-Dimethylphenol	0.53 ± 0.03	0.58 ± 0.01	0.51 ± 0.05	0.56 ± 0.03
6	3-Methylcatechol	2.31 ± 0.06	1.59 ± 0.03	1.83 ± 0.04	1.46 ± 0.07
7	2-Methoxy-4-methylphenol	5.36 ± 0.52	5.13 ± 0.43	5.89 ± 0.19	5.11 ± 0.18
8	4-Methylcatechol	1.39 ± 0.03	2.06 ± 0.04	2.58 ± 0.03	1.55 ± 0.08
9	1,2-Benzenediol	4.58 ± 0.41	4.43 ± 0.19	4.91 ± 0.25	3.93 ± 0.13
10	4-Ethyl-2-methoxyphenol	4.89 ± 0.09	4.97 ± 0.28	3.05 ± 0.05	3.23 ± 0.16
11	4-Methyl-1,2-benzenediol	2.36 ± 0.05	2.68 ± 0.03	2.43 ± 0.07	2.59 ± 0.07
12	2-Methoxy-4-vinylphenol	0.79 ± 0.02	1.25 ± 0.03	1.87 ± 0.02	1.89 ± 0.13
13	2-Methoxy-4-propylphenol	3.58 ± 0.23	4.69 ± 0.19	4.82 ± 0.11	4.56 ± 0.12
Aromatic hydrocarbon					
14	Methylbenzene	1.23 ± 0.03	1.38 ± 0.04	0.76 ± 0.04	0.79 ± 0.02
15	1,4-Dimethylbenzene	1.34 ± 0.08	1.63 ± 0.11	2.06 ± 0.13	1.89 ± 0.13
Aldehydes and ketones					
16	Furfural	5.35 ± 0.13	5.18 ± 0.25	6.29 ± 0.29	6.31 ± 0.27
17	5-Methyl-2-furancarboxaldehyde	0.52 ± 0.01	0.51 ± 0.02	0.36 ± 0.01	0.32 ± 0.01
18	Vanillin	3.25 ± 0.07	3.78 ± 0.08	5.21 ± 0.06	2.59 ± 0.12
19	4-Hydroxy-2-methoxycinnamaldehyde	1.31 ± 0.05	1.23 ± 0.04	0.78 ± 0.05	0.98 ± 0.01
20	Salicylaldehyde	0.58 ± 0.02	0.67 ± 0.03	0.82 ± 0.03	0.93 ± 0.05
21	2-Methyl-2-cyclopenten-1-one	2.31 ± 0.05	2.03 ± 0.08	1.56 ± 0.02	1.05 ± 0.09
22	3-Methyl-1,2-cyclopentanedione	2.31 ± 0.04	2.83 ± 0.12	1.34 ± 0.03	1.59 ± 0.09
23	1-(4-Hydroxy-3-methoxyphenyl)ethanone	0.63 ± 0.02	0.32 ± 0.04	0.25 ± 0.01	0.53 ± 0.02
Acids					
24	Acetic acid	1.82 ± 0.06	1.92 ± 0.08	2.01 ± 0.05	1.32 ± 0.02
25	4-Hydroxybenzoic acid	1.25 ± 0.02	1.37 ± 0.03	0.73 ± 0.02	0.57 ± 0.01

Table 3

Effect of temperature on both the HHV and density of the non-condensable gas.

Temperature (°C)	HHV (MJ/Nm ³)	Density (kg/m ³)	HHV (MJ/kg)
300	5.71 ± 0.23	1.59 ± 0.09	3.57 ± 0.07
400	8.68 ± 0.25	1.43 ± 0.12	6.05 ± 0.09
500	11.85 ± 0.31	1.27 ± 0.11	9.30 ± 0.09
600	12.54 ± 0.46	1.20 ± 0.07	10.49 ± 0.07
700	12.94 ± 0.38	1.13 ± 0.08	11.39 ± 0.06

3. Results and discussion

3.1. Distribution of pyrolysis products

The basic properties of the pine nut shells from the proximate analysis (wt%, db) were found as 80.23% volatile, 19.18% fixed carbon, and 0.59% ash content. The ultimate analysis (wt%, db) revealed the contents of C, H, N, S, and O as 48.26%, 6.43%, 0.82%, 0.41%, and 44.08%, respectively. Both high heating value (19.28 MJ/kg) and the volatile content make pine nut shells a

Table 4

The properties of biochar obtained at different temperatures.

Temperatures (°C)	Proximate analysis (wt%, db)			Ultimate analysis (wt%, db)					HHV (MJ/kg)	Specific surface area (m ² /g)
	Volatile	Fixed carbon	Ash	C	H	N	S	O*		
300	27.58 ± 0.87	71.31 ± 0.87	1.11 ± 0.06	67.23 ± 0.86	4.12 ± 0.13	0.91 ± 0.02	0.45 ± 0.01	26.18 ± 0.87	25.28 ± 0.83	189.2 ± 11.5
400	23.26 ± 0.65	75.53 ± 0.67	1.21 ± 0.18	78.71 ± 0.49	3.76 ± 0.12	0.89 ± 0.02	0.46 ± 0.01	14.97 ± 0.54	30.81 ± 0.86	297.6 ± 11.3
500	19.76 ± 0.43	78.83 ± 0.45	1.41 ± 0.12	85.32 ± 0.93	3.94 ± 0.16	0.85 ± 0.03	0.43 ± 0.01	8.05 ± 0.95	31.76 ± 0.55	356.5 ± 15.2
600	13.32 ± 0.37	85.16 ± 0.38	1.52 ± 0.08	86.34 ± 0.72	3.26 ± 0.08	0.95 ± 0.03	0.39 ± 0.01	7.54 ± 0.73	31.93 ± 0.79	433.1 ± 18.6
700	12.06 ± 0.25	86.21 ± 0.26	1.73 ± 0.07	89.29 ± 0.67	2.34 ± 0.05	0.91 ± 0.02	0.37 ± 0.01	5.36 ± 0.68	32.51 ± 0.23	405.3 ± 19.2

* By difference.

promising raw material of biomass pyrolysis. The effects of pyrolysis temperature on yields of the pyrolysis products are shown in Fig. 1, where both yields of the biochar and bio-oil were obtained by weighing and that of the non-condensable gas was estimated by difference. As the pyrolysis temperature increased, the thermal decomposition of pine nut shells became significant, and more volatiles compounds were released. In turn, this increased yields of both the bio-oil and the non-condensable gas. Elevated temperatures of pyrolysis raised, however, the secondary cracking reactions of the volatiles compounds. This induced a first increase in the bio-oil to reach a maximum yield at 500 °C (43.26%), which slowly reduced afterward. By contrast, the biochar yield continually declined while that of the non-condensable gas increased as the pyrolysis temperature raised. According to the literature, the obtained yields from the pyrolysis of pine nut shells varied according to the experimental conditions. For example, Uçar et al. investigated the pyrolysis of pine nut shells in a fixed bed reactor at 500 °C and estimated both the liquid and solid yields to 45 wt% and 32 wt%, respectively (Uçar and Karagöz, 2014). The variation trends of the products in this study were found in agreement with previously published accounts in the literature (Chen et al., 2015a; Uçar and Karagöz, 2014).

3.2. Properties of pyrolysis products

3.2.1. Bio-oil

The homogeneous bio-oil was analyzed as soon as it was formed. The bio-oil consisted of an aqueous phase and oil phase that underwent a phase separation of bio-oil after several days of storage. The physicochemical properties of the oil phase were then analyzed, and the results are summarized in Table 1. On the other hand, because of the excess water, it was hard to determine the elemental composition of the oil phase and the heating value of the aqueous phase. In the bio-oil, the aqueous phase accounted for 55–85 wt% and the water content was as high as 85–90 wt%. The oil phase, on the other hand, contained a low water content (<10 wt%), elevated carbon content (>57 wt%), and a higher heating value (>22 MJ/kg). As the temperature raised, the yield of the oil phase initially increased due to the induced additional organic components, then followed by a plateau stabilized at about 45 wt%. The water content of both the bio-oil and oil phases presented a downward trend as the temperature increased while the heating value rose. The results from the elemental analysis showed that

elevated pyrolysis temperatures induced carbon contents in the oil phase.

The bio-oil exhibited a very complex composition. The GC/MS analysis of the oil phase showed several chemical compounds, as well as other compounds that could not be detected by GC/MS. The identified compounds of the oil phase with relative peak areas greater than 0.5% were then analyzed, and the results were presented in Table 2. The oil phase comprised mainly phenolic compounds, including phenol, 2-methoxyphenol, and 1,2-benzenediol with elevated relative peak areas. Other high added-value compounds, such as vanillin and aromatic hydrocarbons, were also present in the oil phase, which may promote the application of the produced bio-oil.

The pyrolysis mechanism of the biomass was associated with the cellulose, hemicelluloses, and lignin. At temperatures above 300 °C, the pyrolysis of cellulose involved mainly two major parallel pathways related to depolymerization and fragmentation of the ring scission, which generated the liquid products at very high yields. The depolymerization process induced mainly levoglucosan (LG), furans, cyclopentanones, pyrans, and other related derivatives. The ring scission process mostly formed hydroxyacetaldehyde (HAA), acetol (HA), and other linear alcohols and esters. If compared to cellulose, hemicelluloses were thermally less stable and their pyrolysis reaction mechanisms had some similarities to those of cellulose. In general, xylan pyrolysis induced mainly water, furfural, and acids (formic acid and acetic acid) but typical products like LG from cellulose were not formed. On the other hand, lignin was the most complex and greatest thermally stable component of biomass. The details of possible routes for the pyrolytic formation of the main products from cellulose and xylan have been discussed in the literature (Shen and Gu, 2009; Shen et al., 2010). In general, lignin pyrolysis was produced with higher char yield and lower liquid yield than holocellulose, where the liquid product contained mainly larger molecular oligomers known as pyrolytic lignins, phenol based-compounds, and lighter compounds (methanol, HAA, acetic acid) (Zhu and Lu, 2013).

3.2.2. Non-condensable gas

Fig. 2 depicts the effect of the pyrolysis temperature on the formed gas components of the non-condensable gas. Both CO₂ and CO were observed as the dominant gases but their volume fractions reduced rapidly as the pyrolysis temperature increased. On the other hand, the volume fractions of CH₄ and H₂ notably increased with the temperature. The previous study has shown

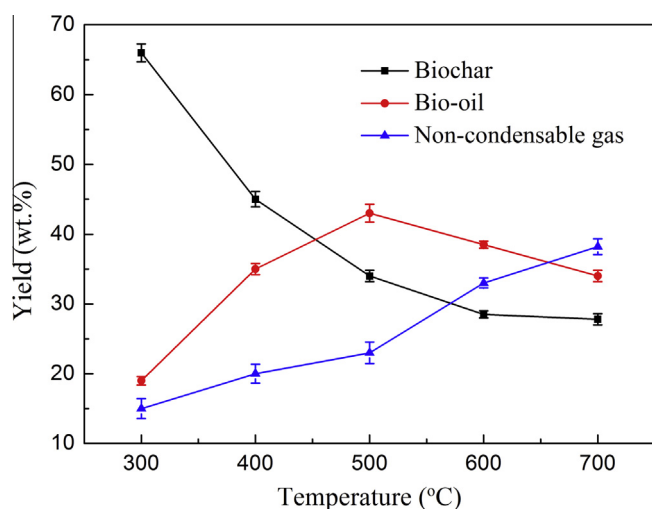


Fig. 1. Effect of temperature on yields of the pyrolysis products of pine nut shells.

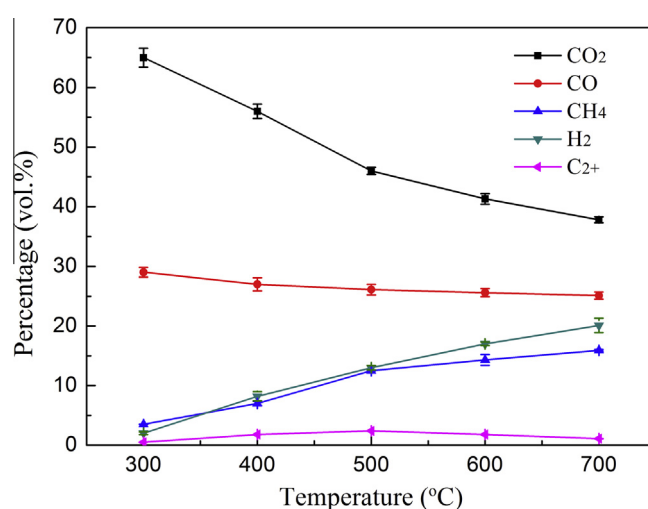


Fig. 2. Effect of temperature on gas components of the non-condensable gas.

that CO₂ and CO were primarily formed during the early stages of the pyrolysis process resulting mainly from the cellulose and hemicellulose decomposition, while the amount of H₂ and CH₄ gradually increased as a consequence of the slow decomposition of lignin (Chen et al., 2014a). Another study has demonstrated that the generated CO and CO₂ at temperatures below 600 °C was generally the result of cracking-reforming processes of functional groups related to carboxyl (C=O), carbonyl (C–O–C), and carboxylic acid (COOH) (Yang et al., 2007). At temperatures above 600 °C, the release of CO and CO₂ is likely to be induced by the secondary thermal cracking of the bio-oil. Other gases, including C₂H₂, C₂H₄, C₂H₆, C₃H₆, and C₃H₈, were also found in the products but at very low volume fractions.

The HHV of the non-condensable gas at different units can be obtained after calculation of density of the non-condensable gas (Table 3). The heating values of H₂ and CH₄ were much higher (141.8 MJ/kg for H₂, 55.5 MJ/kg for CH₄) than that of CO (10.1 MJ/kg). Therefore, the heating value of the non-condensable gas directly increased from 3.57 MJ/kg (5.71 MJ/Nm³) at 300 °C to 11.39 MJ/kg (12.94 MJ/Nm³) at 700 °C, as the volume fractions of H₂ and CH₄ raised with the temperature.

3.2.3. Biochar

The properties of the biochar listed in Table 4 revealed that the increase in pyrolysis temperature from 300 °C to 700 °C induced intense reactions of cellulose, hemicellulose, and lignin. In turn, this released larger amounts of volatile matter, which declined the yields and volatile contents of the biochar. Furthermore, the inorganic components present in pine nut shells were nearly preserved in the biochar during the thermal decomposition process, leading to the gradual increase in the ash content of biochar. Meanwhile, boosting the pyrolysis temperature raised the heating value of biochar. Fig. 1 clearly depicts that the yield of biochar was significantly reduced at temperatures below 500 °C then continued to decrease but slowly when the temperatures were above 500 °C. The latter was because most of the cellulose, hemicellulose, and lignin underwent decomposition at temperatures below 500 °C, and the pyrolysis of pine nut shells above 500 °C involved the slow decomposition of the remaining residues that induced only slight weight losses. During the pyrolysis process, the original biomass structure was destroyed to yield formation of porous structure. The specific surface area, which directly reflects the adsorption capacity of the biochar, was closely associated with the pore size. Table 4 clearly depicts that the specific surface area increased rapidly during the initial stages to reach a maximum value of 433.1 m²/g at 600 °C. This could be attributed to the release of volatile compounds at temperatures ranging from 300 °C to 600 °C, which forced openings for gas escape and formed pores in biochar. Nevertheless, a further increase in the pyrolysis temperature reduced the specific surface area that might be related to melting of the ash content, which blocked part of the formed pores by tar (Chen et al., 2012).

3.3. Pyrolysis polygeneration of pine nut shells

3.3.1. Energy distribution of pyrolysis products

Fig. 3 illustrates the energy distribution of pyrolysis products of the pine nut shells. The total energy yield of the three-phase products exhibited a deviation of ±4% from 100%, possibly due to errors made during collection of the products, as well as the calculation of yields and densities of the non-condensable gas. The energy yield of the bio-oil showed an initial increase from 7.17% at 300 °C to 35.28% at 500 °C followed by a decline to about 28.05% at 700 °C. Overall, the energy yields of the bio-oil were higher than those of the non-condensable gas but lower than those of the biochar. On the other hand, although the mass yields of the biochar were infe-

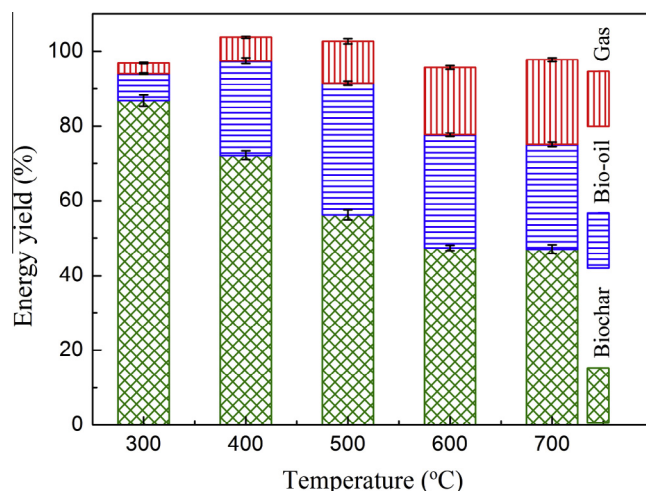


Fig. 3. Effect of temperature on the energy yield of pyrolysis products.

rior to those of non-condensable gas and bio-oil, the energy yields of biochar were 47.05–86.85% superior, which were significantly higher than those related to other two-phase products. As a result, biochar was considered as the primary product obtained from the pyrolysis polygeneration of pine nut shells in terms of mass and energy yields.

3.3.2. Further utilization of pyrolysis products

The non-condensable gas contained combustible components, such as CO, H₂, CH₄, and C_nH_m. The gas contributed to its elevated heating value that reached amounts up to 12.94 MJ/m³ at 700 °C. Thus, a non-condensable gas possibly underwent combustion in turbines and boilers. Also, it can be used for supplying heat to pyrolysis equipment to achieve energy self-sufficiency (Chen et al., 2015b; Kovacs and Meggyes, 2009). The bio-oil from pine nut shells pyrolysis contained various ingredients that could directly be employed as fuel oil or be refined to a high-quality liquid fuel (Bridgwater, 2012; Kim et al., 2014; Uçar and Karagöz, 2014). Furthermore, according to the characteristics of the oil phase, pyrolysis of pine nut shells may be utilized as a useful tool for producing valuable chemicals or chemical feedstock for plastics, pesticides, pharmaceuticals, and dyes (Chen et al., 2014b; Kim et al., 2014). In turn, biochar can be converted into organic fertilizers, fast-burning carbon, and activated carbon widely used in agriculture, industrial production, and daily life (Kambo and Dutta, 2015; Qian et al., 2015; Shim et al., 2015). The contents of cellulose, hemicellulose, and lignin were determined as 36.13%, 21.97%, and 39.59%, respectively. Lignin was the primary carbon-forming material of biomass during the pyrolysis process, and if compared with cotton stalk, rapeseed stalk, tobacco stem, rice husk and bamboo, pine nut shells exhibited higher lignin content (Chen et al., 2016). Hence, preparing activated carbon using pine nut shells is one proper pathway for utilizing resources.

The above-discussed results indicated that pyrolysis temperatures of 500–600 °C generated products with superior heating values of non-condensable gas, lower water contents, high heating values of bio-oil, elevated fixed carbon content, and greater specific surface areas of biochar. Thus, this temperature range would probably be favorable for a decent utilization of all products. On the other hand, this temperature range was also required in physical activation method for preparation of activated carbon from biomass (wood, charcoal, nut shell). The physical activation method consisted of two steps: i) preparation of biochar or carbonized material through carbonization or pyrolysis at around 600 °C, and ii) activation of biochar for the preparation of activated carbon

at higher temperatures. Here, the biochar obtained from pyrolysis at 600 °C was used as raw material for the production of activated carbon.

3.4. Preparation of activated carbon

Activated carbon has very excellent adsorbent, where steam is one of the most common activators used by physical methods due to its numerous advantages, including the environment-friendly and fewer post-processing required preparation steps. The utilized activation temperature was usually around 800–900 °C, under which carbon atoms reacted with the activator (steam) to form CO, H₂, and CO₂, as well as to develop micropores within the material. The activation reaction occurred in the gas-solid phase system of multiphase reaction, where steam first reacted with disordered carbon atoms and impurity atoms to open the blocked pores then further widening of the pores developed to form new ones. The main chemical reactions were summarized as follows: $C + H_2O \rightarrow H_2 + CO$, $C + H_2O \rightarrow H_2 + CO_2$, $C + H_2 \rightarrow CH_4$, and $C + CO_2 \rightarrow CO$. The activation conditions, including the temperature, time, and activator flow (steam/biochar ratio) had important roles in the quality of the resulting activated carbon.

3.4.1. Effect of activation temperature

It is imperative to investigate activation temperature as it significantly affects the preparation of activated carbon. Fig. 4 shows the effect of activation temperature (750–950 °C) on the yield and adsorption performance of the activated carbon under activation time of 60 min and steam/biochar ratio of 2.

It can be observed that as the activation temperature got higher, the yield of activated carbon decreased from 56.21% to 23.16%. The latter might be attributed to the fact that the increase in temperature induced a more severe reaction between carbon and steam that, in turn, raised the number of the activated carbon atoms. Thus, a series of reactions occurred, resulting in high porosity activated carbon but with moderate yield.

There was a constant increase of iodine adsorption capacity of activated carbon with temperature up to 850 °C then started to decline afterward. The specific surface area of activated carbon presented a variation trend similar to that of the iodine adsorption capacity. As the temperature incremented up to 850 °C, there was a constant reaction of the steam with carbon atoms at the reaction

sites that generated abundant microporous and some mesopores structures. Thus, the iodine adsorption capacity constantly increased. At 850 °C, the activated carbon exhibited the optimum adsorption performance regarding the yield (36.11%), iodine value (890.2 mg/g), and specific surface area for the activated carbon (1016.7 m²/g). At temperatures above 850 °C, the micropores were constantly transformed into mesopores or even macropores, resulting in low values of both the iodine adsorption capacity and specific surface area of the activated carbon.

3.4.2. Effect of activation time

Fig. 5 shows the effect of activation time (40–120 min) on the yield and adsorption performance of activated carbon at the activation temperature of 850 °C and steam/biochar ratio of 2. At shorter activation periods, the bio-char obtained from pyrolysis polygeneration of pine nut shells did not have sufficient time to produce well-developed pores, hence resulting in moderate adsorption capacities. As the activation time is increased, the yields of activated carbon constantly reduced. This indicated that prolonged activation times resulted in the elevated extent of activation and greater reaction between the steam and carbon. Simultaneously, longer activation periods allowed a constant penetration of the steam into the obtained biochar that, in turn, accelerated desorption, distillation, and removal of volatile compounds. As a result, increased activation time induced elevated iodine adsorption values of activated carbon. When the activation time was beyond 80 min, a reduction trend was noticed probably due to the collapse and loss in numbers of the formed pores. On the other hand, the specific surface area of activated carbon represented a variation trend similar to that of the iodine adsorption capacity. The optimum adsorption performance was observed at 80 min activation time, which induced a yield of 25.78%, an iodine value of 961.2 mg/g, and specific surface area of activated carbon of 1036.5 m²/g.

3.4.3. Effect of steam/biochar ratio

The effect of steam/biochar ratios ranging from 0.5 to 2.5 on the yield and adsorption performance of activated carbon were studied at 850 °C activation temperature and 80 min activation time (Fig. 6). The increase in steam/biochar ratio reduced the yield of activated carbon. In other words, incremented steam/biochar ratios accelerated the reaction of carbon with steam, resulting in

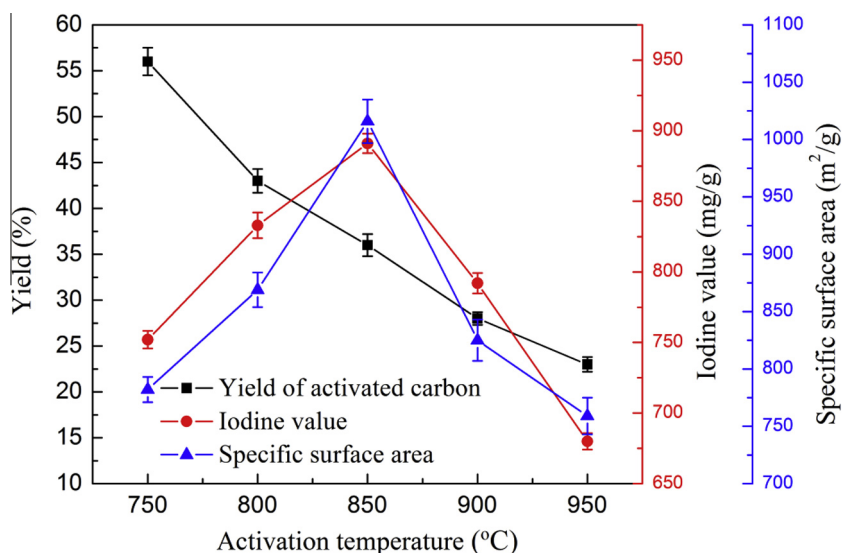


Fig. 4. Effect of activation temperature on the yield and adsorption performance of activated carbon.

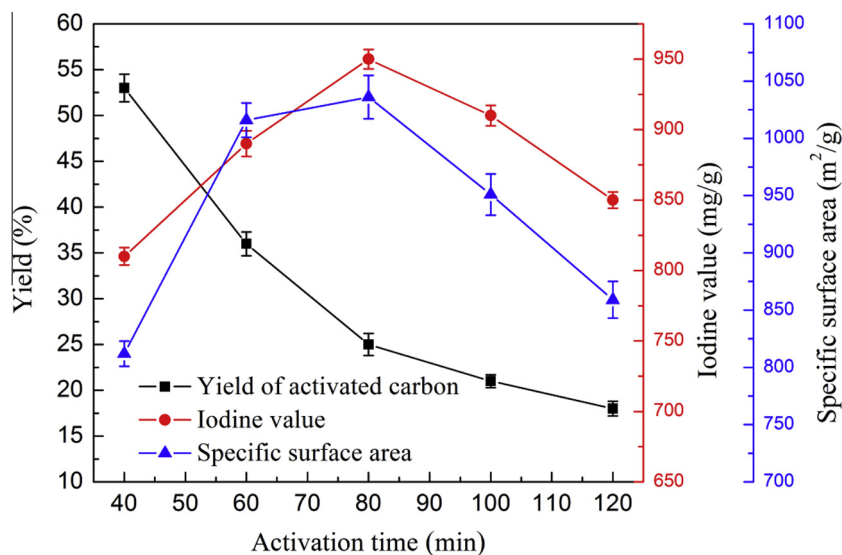


Fig. 5. Effect of activation time on the yield and adsorption performance of activated carbon.

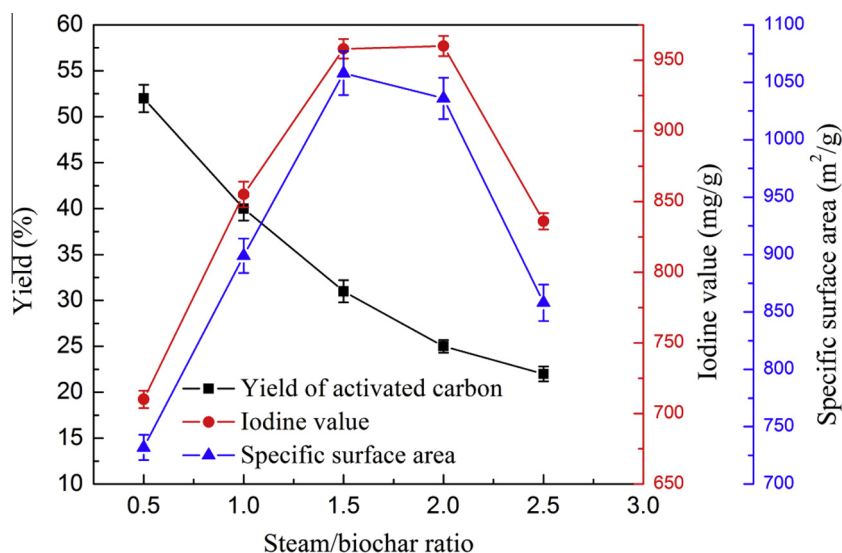


Fig. 6. Effect of steam/biochar ratio on the yield and adsorption performance of the activated carbon.

lower yields of activated carbon. The iodine adsorption followed a similar trend of variation as the specific surface area. The increase in steam/biochar ratio accelerated the activation reaction rate that facilitated the formation of abundant micropores and several mesopores, resulting in raised iodine adsorption capacities. Higher steam/biochar ratios above 1.5 generated more mesopores and macroporous formed from micropores. On the other hand, elevated water flows entering the activation tube caused a loss of heat that was absorbed by the activation reaction, thereby declining the surface temperature of the activated carbon. Hence, the reaction between steam and carbon became slow and prevent the formation of new micropores, resulting in lower iodine adsorption capacities. As the steam/biochar ratio reached 1.5, the specific surface area exhibited a declining trend.

The optimal activation conditions were overall determined to be: 850 °C activation temperature, 80 min activation time, and steam/biochar ratio of 1.5. Under these circumstances, the activated carbon with acceptable yield and good adsorption capacity was obtained.

4. Conclusions

With increasing pyrolysis temperature, the yield of biochar decreased while that of bio-oil increased initially then followed by decline. Biochar was found to be the primary product issued from the pyrolysis reaction with high energy yields, and 600 °C was recorded as the optimal temperature for the production of biochar. For activated carbon, the optimal activation conditions were found to be 850 °C activation temperature, 80 min activation time, and a steam/biochar ratio of 1.5. Under these conditions, the activated carbon yield, the iodine adsorption value, and the specific surface area were 31.2%, 958.9 mg/g, and 1057.8 m²/g, respectively.

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